

# QMS Journal Club

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## QMS Journal Club 3rd meeting: Postulates of quantum mechanics and Bloch sphere II

# Phase

Global phase factor is a term  $e^{i\theta}$  attached to  $|\psi\rangle$  without changing the measurement statistics of  $|\psi\rangle$ ,

$$\langle\psi|e^{-i\theta}M_m^\dagger M_me^{i\theta}|\psi\rangle = \langle\psi|M_m^\dagger M_m|\psi\rangle. \quad (1)$$

For this reason, we say that  $|\psi\rangle \sim e^{i\theta}|\psi\rangle$ .

Relative phase is a term  $e^{i\theta}$  attached between probability amplitudes,

$$|\psi\rangle = \alpha|\psi_\alpha\rangle + e^{i\theta}\beta|\psi_\beta\rangle. \quad (2)$$

Relative phase is a basis-dependent concept and may give rise to physically observable differences in measurement statistics.

For example,  $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$  and  $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$  are not the same up to a relative phase shift.

For one quantum bit (qubit), the Hilbert space is given as  $\mathcal{H} = \mathbb{C}^2$ , where

$$|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \quad (3)$$

$$= \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}, \quad (4)$$

where  $\psi_0$  and  $\psi_1$  are complex numbers, and  $|\psi_0|^2 + |\psi_1|^2 = 1$ .

At first glance, there are 4 degrees of freedom. However, due to the normalization condition and the equivalence of global phase factor, we only need 2 degrees of freedom to parametrize a one-qubit state,

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle. \quad (5)$$

Given the following Pauli matrices, find the expected values of Pauli- $X$ ,  $Y$ , and  $Z$  for  $|\psi\rangle$ .

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (6)$$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle. \quad (7)$$

$$\langle\psi|X|\psi\rangle = \quad (8)$$

$$\langle\psi|Y|\psi\rangle = \quad (9)$$

$$\langle\psi|Z|\psi\rangle = \quad (10)$$

The expected values of Pauli matrices correspond to the spherical coordinates when the radius equals to 1,

$$\langle \psi | X | \psi \rangle = \sin \theta \cos \phi, \quad (11)$$

$$\langle \psi | Y | \psi \rangle = \sin \theta \sin \phi, \quad (12)$$

$$\langle \psi | Z | \psi \rangle = \cos \theta. \quad (13)$$

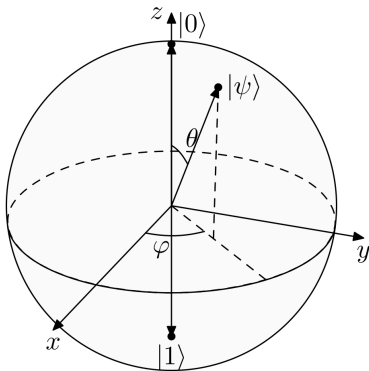


Figure 1: Bloch sphere

# Postulates of quantum mechanics

Postulate 4: The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through  $n$ , and system number  $i$  is prepared in the state  $|\psi_i\rangle$ , then the joint state of the total system is

$$|\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_n\rangle. \quad (14)$$

Postulate 4 enables us to define quantum entanglement. Suppose we are given two qubit states,  $|\psi\rangle \in \mathcal{H}_A = \mathbb{C}^2$  and  $|\phi\rangle \in \mathcal{H}_B = \mathbb{C}^2$ . The two-qubit state  $|\psi_{AB}\rangle$  is entangled if  $|\psi_{AB}\rangle \neq |\psi\rangle \otimes |\phi\rangle$ .

For example, the following state  $|\psi_{AB}\rangle$  is separable,

$$\begin{aligned} |\psi_{AB}\rangle &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \\ &= \frac{1}{2}[|0\rangle \otimes (|0\rangle + |1\rangle) + |1\rangle \otimes (|0\rangle + |1\rangle)] \\ &= \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \\ &= |+\rangle \otimes |+\rangle \end{aligned}$$

It is not always obvious to identify if a given two-qubit state is separable or not. One can derive separability condition for two qubits. As long as the two-qubit state satisfies the separability condition, the two-qubit state is separable (else, it is entangled).

Let  $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$ ,  $|\phi\rangle = \phi_0|0\rangle + \phi_1|1\rangle$ . Then

$$\begin{aligned} |\psi_{AB}\rangle &= |\psi\rangle \otimes |\phi\rangle \\ &= (\psi_0|0\rangle + \psi_1|1\rangle) \otimes (\phi_0|0\rangle + \phi_1|1\rangle) \\ &= \psi_0\phi_0|00\rangle + \psi_0\phi_1|01\rangle + \psi_1\phi_0|10\rangle + \psi_1\phi_1|11\rangle \end{aligned}$$

Notice that the probability amplitudes of  $|00\rangle$  and  $|11\rangle$ , when multiplied together, is the same as the product of probability amplitudes of  $|10\rangle$  and  $|01\rangle$ . We can conclude that for a two-qubit state  $|\psi_{AB}\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle$ , if  $\psi_{00}\psi_{11} = \psi_{10}\psi_{01}$ , the two-qubit state is separable. Else, it is entangled.

Can Alice communicate a number of classical bits of information by transmitting a smaller number of qubits, under the assumption that Alice and Bob receive an entangled resource beforehand?

- Alice and Bob initially share a pair of qubits in the Bell state,

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}; \quad (15)$$

- If Alice wants to send 00, she does nothing; If Alice wants to send 01, she applies  $Z$ ; If Alice wants to send 10, she applies  $X$ ; If Alice wants to send 11, she applies  $iY$ ;

- The final two-qubit state becomes:

$$00 : |\psi\rangle \rightarrow \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (16)$$

$$01 : |\psi\rangle \rightarrow \frac{|00\rangle - |11\rangle}{\sqrt{2}} \quad (17)$$

$$10 : |\psi\rangle \rightarrow \frac{|10\rangle + |01\rangle}{\sqrt{2}} \quad (18)$$

$$11 : |\psi\rangle \rightarrow \frac{|01\rangle - |10\rangle}{\sqrt{2}} \quad (19)$$

- Alice sends her qubit to Bob;
- Bob performs a Bell basis measurement to determine which of the four possible bit strings Alice sent.

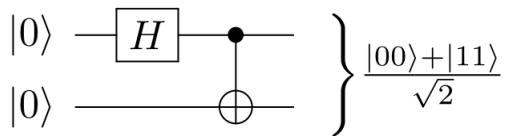


Figure 2: Bell state creation

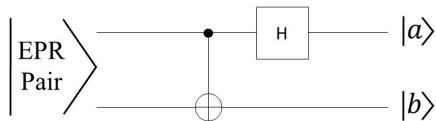


Figure 3: Bell basis measurement